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SYSTEMS AND METHODS FOR DEPLOYING AND SECURING ENDOTRACHEAL TUBES

Abstract

An endotracheal tube can include a tube having a distal end that is configured to be inserted into the trachea of a patient and a proximal end that is configured to be positioned within the mouth of the patient or outside of the patient. The endotracheal tube can include a tether having a first end coupled to the tube, and a magnet coupled to the tether. The magnet can be configured to be advanced through the pharynx, through and out the nasal cavity thereby positioning the soft palate of the patient between the tube and the tether.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of and priority to U.S. Provisional Application Ser. No. 63/328,946, filed on Apr. 8, 2022, the entirety of which is incorporated by reference herein.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH
[0002] N/A.
BACKGROUND
[0003] Endotracheal tubes are typically deployed in medical situations in which the patient is unable to breathe (or is having difficulty breathing) on his/her own, which can occur during general anesthesia procedures in which anesthesia temporarily paralyzes the breathing muscles including the diaphragm, or during other scenarios in which the patient is having difficulty breathing (e.g., due to respiratory diseases, pneumothorax, heart failure, overdose, stroke, etc.), etc.
[0004] While the use of endotracheal tubes can be extremely helpful in establishing and maintaining a patient's airway, in some cases, patients can inadvertently remove the endotracheal tube, thereby compromising the established airway. For example, after waking from a sedated state, a confused and startled patient may improperly believe that the endotracheal tube is choking them, when in fact, the endotracheal tube is allowing the patient to breathe. Thus, such a patient may tug at the endotracheal tube with the aim of removing the tube, which can not only dislodge the endotracheal tube (e.g., requiring redeployment of the endotracheal tube, which undesirably cause patient discomfort and consume additional hospital resources including materials and practitioner expertise), but can even cause death if, for example, the airway is unable to be reestablished within a certain amount of time.
[0005] Thus, it would be desirable to have improved systems and methods for deploying and securing endotracheal tubes.
SUMMARY OF THE DISCLOSURE
[0006] Some embodiments of the disclosure provide an endotracheal tube. The endotracheal tube may include a tube having a distal end that is configured to be inserted into the trachea of a patient and a proximal end that is configured to be positioned within the mouth of the patient or outside of the patient; a tether having a first end coupled to the tube; and a magnet coupled to the tether, the magnet being configured to be advanced through the pharynx, through and out the nasal cavity thereby positioning the soft palate of the patient between the tube and the tether.
[0007] In certain embodiments of the endotracheal tube, when the proximal end of the tube is advanced away from the patient, the tether may contact the soft palate to block further advancement of the proximal end of the tube away from the patient.
[0008] In various embodiments, the endotracheal tube may further include a channel that extends along the tube; a slide coupled to a second end of the tether opposite the first end; and a stop coupled to the tube, where the magnet may be configured to be moved along the channel in a first direction towards the proximal end of the tube until the slide contacts the stop thereby blocking further advancement of the magnet in the first direction.
[0009] In particular embodiments of the endotracheal tube, a second end of the tether opposite the first end may be configured to be coupled to an anchor that is external to the patient thereby fixing the position of the endotracheal tube within the patient.
[0010] In some embodiments of the endotracheal tube, the anchor may wrap around the head of the

patient, and a portion of the tube may be coupled to the anchor.
[0011] In certain embodiments of the endotracheal tube, the magnet may be removably coupled to the tether.
[0012] Particular embodiments of the endotracheal tube may further include a first adapter coupled to the proximal end of the tube, a second adapter of a conduit in fluid communication with a ventilator may be configured to be removably coupled to the first adapter, and the first adapter may be configured to be positioned within the mouth of the patient when the distal end of the tube is placed in the trachea of the patient.
[0013] In various embodiments of the endotracheal tube, when the second adapter is pulled away from the first adapter of the tube, the second adapter may decouple from the first adapter.
[0014] In some embodiments of the endotracheal tube, the magnet may be a first magnet, the first adapter may include a second magnet, the second adapter may include a third magnet, and the first adapter may magnetically couple to the second adapter, via the interaction between the second magnet and the third magnet.
[0015] Certain embodiments of the endotracheal tube may further include an inflatable cuff coupled to the tube which may be configured to be positioned within the trachea of the patient, and when the first adapter is coupled to second adapter and the second adapter is pulled away from the first adapter, the second adapter may decouple from the first adapter with the distal end of the tube remaining in the trachea.
[0016] In various embodiments of the endotracheal tube, the magnet may be a first magnet, and a probe with a second magnet coupled thereto may be configured to be inserted into the nasal cavity, through the nasal cavity, into the pharynx to be magnetically coupled to the first magnet and, with the second magnet magnetically coupled to the first magnet, the probe may be configured to be retreated back through and out the nasal cavity.
[0017] Some embodiments of the disclosure provide a method for deploying an endotracheal tube. The methods may include inserting a distal end of a tube into the trachea of a patient; positioning a proximal end of the tube within the mouth of the patient or outside of the patient, a first end of a tether being coupled to the tube, and a magnet being coupled to the tether; and advancing the magnet through the pharynx of the patient, through and out the nasal cavity such that the soft palate of the patient is positioned between the tube and the tether.
[0018] In certain embodiments of the method, when the proximal end of the tube is advanced away from the patient, the method may further include contacting the soft palate with the tether to block further advancement of the proximal end of the tube away from the patient.
[0019] Various embodiments of the method may further include: providing a channel that extends along the tube, where a slide may be coupled to a second end of the tether opposite the first end and a stop may be coupled to the tube; and moving the magnet along the channel in a first direction towards the proximal end of the tube until the slide contacts the stop to block further advancement of the magnet in the first direction.
[0020] In particular embodiments, the method may further include coupling a second end of the tether opposite the first end to an anchor that is external to the patient to fix the position of the endotracheal tube within the patient.
[0021] In some embodiments, the method may further include wrapping the anchor around the head of the patient, and coupling a portion of the tube to the anchor.
[0022] In various embodiments of the method, the magnet may be removably coupled to the tether.
[0023] Certain embodiments of the method may further include: providing a first adapter coupled to the proximal end of the tube, where a second adapter of a conduit in fluid communication with a ventilator may be configured to be removably coupled to the first adapter; and positioning the first adapter within the mouth of the patient when the distal end of the tube is placed in the trachea of the patient.
[0024] In particular embodiments of the method, when the second adapter is pulled away from the

first adapter of the tube, the method may further include decoupling the second adapter from the first adapter.
[0025] In some embodiments of the method, the magnet may be a first magnet, wherein the first adapter may include a second magnet, the second adapter may include a third magnet, where the method may further include magnetically coupling the first adapter to the second adapter based on the interaction between the second magnet and the third magnet.
[0026] Various embodiments of the method may further include: providing an inflatable cuff coupled to the tube, positioning the inflatable cuff within the trachea of the patient, coupling the first adapter to the second adapter, and decoupling, when the second adapter is pulled away from the first adapter, the second adapter from the first adapter, wherein the distal end of the tube may remain in the trachea.
[0027] In particular embodiments of the method, the magnet may be a first magnet and the method may further include: inserting a probe with a second magnet coupled thereto into the nasal cavity, through the nasal cavity, and into the pharynx, magnetically coupling the second magnet to the first magnet, and retreating the probe with the second magnet magnetically coupled to the first magnet back through and out the nasal cavity.
[0028] The foregoing and other aspects and advantages of the present disclosure will appear from the following description. In the description, reference is made to the accompanying drawings that form a part hereof, and in which there is shown by way of illustration one or more exemplary versions. These versions do not necessarily represent the full scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0029] The following drawings are provided to help illustrate various features of non-limiting examples of the disclosure, and are not intended to limit the scope of the disclosure or exclude alternative implementations.
[0030] FIG. **1** shows a schematic illustration of an endotracheal tube.
[0031] FIG. **2** shows a probe that is configured to retrieve the magnet of the endotracheal tube of FIG. **1** depicted in a straight configuration.
[0032] FIG. **3** shows the probe of FIG. **2** depicted in a curved configuration.
[0033] FIG. **4** shows a schematic illustration of another endotracheal tube.
[0034] FIG. **5** shows a schematic illustration of the endotracheal tube of FIG. **1** deployed within the patient.
[0035] FIG. **6** shows a schematic illustration of an endotracheal tube system with adapters of the endotracheal tube system separated from each other.
[0036] FIG. **7** shows a schematic illustration of the endotracheal tube system of FIG. **7** with the adapters coupled together.
[0037] FIG. **8** shows a schematic illustration of another endotracheal tube system.
[0038] FIG. **9** shows a schematic illustration of another endotracheal tube system.
DETAILED DESCRIPTION OF THE PRESENT DISCLOSURE
[0039] As described above, a patient may inadvertently remove a previously deployed endotracheal tube (i.e., an unplanned extubation). Unplanned extubations occur frequently in the medical setting, and have high mortality risks associated therewith. However, there are few (if any) devices available that can mitigate, or deter removal of, a deployed endotracheal tube to prevent an unplanned extubation and many that are in existence rely on affixing an appliance to the skin and connecting/adhering this appliance to the outer lumen of the endotracheal tube
[0040] Some embodiments of the disclosure provide advantages for addressing these and other issues by providing improved systems and methods for deploying and securing endotracheal tubes.

For example, some embodiments of the disclosure provide an endotracheal tube that can behaviorally deter removal of the endotracheal tube (e.g., by the patient with the endotracheal tube deployed therein), and can further secure the endotracheal tube relative to the patient, both of which can decrease the likelihood of an unplanned extubation (e.g., in an intensive care unit setting or in the operating room where mechanical forces are inadvertently applied to the tube directly through movement of the patient thereby leading to removal of the endotracheal tube. As a more specific example, the endotracheal tube can include a tube, a tether with one end coupled to the tube, and a magnet coupled to the tether. After the endotracheal tube is positioned in the trachea of the patient, a (flexible) probe with a magnet coupled thereto can be inserted into the nasal cavity, through the nasal cavity, into the pharynx, and the magnet of the probe can be magnetically coupled to the magnet of the endotracheal tube. Then, the probe with the magnets coupled together, can be retracted back through the nasal cavity and out of the patient, and the tether can be secured to the patient (e.g., an anchor, such as, for example, a tube holder). In this way, because the tether is coupled to the endotracheal tube, securing the tether outside of the patient further secures the endotracheal tube (e.g., by blocking pulling of the endotracheal tube out of the patient), while deterring the patient from removing the endotracheal tube. For example, when the endotracheal tube is deployed, the tube is positioned within the trachea, while the tether extends along the pharynx, through the nasal cavity, and out of the nasal cavity. Thus, the soft and hard palate are positioned between the tether and the tube. In this way, if the endotracheal tube is pulled away from the patient, the tether contacts the soft palate, which deters the patient from pulling the endotracheal tube further away (e.g., because this can be uncomfortable, or can cause pain and the hard palate is rigid and will provide a fixation point for the tethering apparatus described above).

[0041] FIG. **1** shows a schematic illustration of an endotracheal tube **100**. The endotracheal tube **100** can include a tube **102** having a proximal end **104** and a distal end **106** opposite the proximal end **104**, an inflatable cuff **108** coupled to the tube **102**, a pilot balloon **110** in fluid communication with the inflatable cuff **108**, a tether **112**, a magnet **114**, a slide **116**, a stop **118**, and a channel **120**. In some embodiments, the distal end **106** of the tube **102** can be inserted into the mouth of the patient, and advanced until the distal end **106** reaches the trachea of the patient. At this point, the proximal end **104** of the tube **102** can be positioned outside of the patient, or within the mouth of the patient. As shown in FIG. **1**, the inflatable cuff **108** can be coupled to the tube **102** at (or proximal to) the distal end **106** of the tube **102**. In this way, when the inflatable cuff **108** is inflated, the inflatable cuff **108** contacts the trachea to secure the tube **102** (and in particular the distal end **106** of the tube **102**) within the trachea. In some cases, the inflatable cuff **108** can extend entirely around the tube **102**, and the inflatable cuff **108** can be selectively inflated via a fluid source (e.g., a pump, such as an air pump). For example, fluid from a fluid source can be driven through the pilot balloon **110** and into the inflatable cuff **108** to inflate the inflatable cuff **108**. In some cases, the pilot balloon **110** can provide a visual indication as to the degree to which the inflatable cuff **108** is inflated. For example, the greater the inflation of the pilot balloon **110**, the greater the inflation of the inflatable cuff **108**.

[0042] As shown in FIG. **1**, the magnet **114** is coupled to the tether **112**, and the slide **116** is coupled to the tether **112**. In particular, the magnet **114** is coupled to one end of the tether **112** and the slide **116** is coupled to the other end of the tether **112**. In other configurations, however, the magnet **114** and the slide **116** can be coupled to other portions of the tether **112**. For example, the magnet **114** can be coupled to the tether **112** away from either end of the tether **112** (e.g., centrally located on the tether **112**), while the slide **116** can be coupled to the tether **112** away from the other end of the tether **112**. Regardless of the configuration, the magnet **114** and the slide **116** can be coupled to the tether **112** so that the magnet **114** is separated from the slide **116**. As shown in FIG. **1**, the tether **112** with the magnet **114** and the slide **116** coupled thereto, can be positioned within a channel **120**. In some cases, the wall that defines the channel **120** can be coupled to the outside of the tube **102**, or inside of the tube **102** (e.g., within the lumen of the tube **102**). In other cases, the channel **120** can be directed through a wall of the tube **102**. In other cases, the endotracheal tube **100** does not include the channel **120**, and in other cases, the magnet **114** can be positioned outside of the channel **120** with the slide **116** positioned within the channel **120**.

[0043] The stop **118** serves to hold the slide **116** onto the tube **102** (e.g., at the end of the channel **120**) in order to anchor the tether **112** to the tube **102**. In some embodiments, the stop **118** can be coupled to the tube **102** and can be positioned at one end of the channel **120**, which can be closer to the distal end **106** of the tube **102**. For example, depending on the anchor location on the tube **102**, the position of the stop **118** can be moved to, for example, accommodate differences in anatomical features (e.g., the size of the mouth). In this regard, for example, the stop **118** can be slideably engaged within the channel **120** to adjust the position of the stop **118** relative to the tube **102** to accommodate for differing anatomies (e.g., to ensure that the anchor point is within the trachea).

[0044] The stop **118** can be implemented in different ways. For example, the stop **118** can be a hole, a protrusion, etc. Regardless of the configuration, when the magnet **114** is pulled (e.g., by another magnet), the magnet **114** can be pulled in an opposing direction, until the slide **116** reaches and contacts the stop **118**. At this point, further puling of the magnet **114** will be blocked due to the stop **118** contacting and thus blocking movement of the slide **116** in the direction of the pulling. For example, as the magnet **114** is pulled, the tether **112** passes through (or by) the stop **118** (e.g., due to the size of tether being smaller than the stop **118**), however, the slide **116** is larger than the stop **118**, and thus the stop **118** is unable to move past the stop **118** in the direction opposite to the pulling direction.

[0045] The magnet **114** can be implemented in different ways. For example, the magnet **114** can be a single magnet or a group of magnets. For example, the magnet **114** can be an array of individual magnets that are coupled together (e.g., a row of magnets, a 2-D array of magnets, etc.). In this way, a probe that includes a magnet can more easily locate the array of magnets that make up the magnet **114** at least because the array of magnets takes up a larger extent of the tube **102**, and can provide a greater attractive force to magnetically attract a magnet that is associated with the probe. In some cases, the magnet **114** can be a permanent magnet, such as, for example, a rare-earth magnet (e.g., a neodymium magnet). The slide **116** can also be implemented in different ways. For example, the slide **116** can have different shapes (e.g., a prism, a sphere, an ellipse, etc.). In some cases, the slide **116** having one or more curves (e.g., the slide **116** being a sphere) can be advantageous in that the slide **116** can more easily slide within the channel **120**, and the slide **116** is less likely to undesirably puncture the wall of the channel **120** when the slide **116** has one or more curves. In some cases, the shape of the slide **116** can be complementary to the shape of the stop **118**. For example, the slide **116** can be convex (or concave), while the stop **118** can be concave (or convex). In this way, the slide **116** can more easily interface with the stop **118**.

[0046] In some embodiments, the magnet **114** can be removably coupled to the tether **112**, or can be removable from the tether **112**. For example, the tether **112** can include a region of material weakness (e.g., a perforation, a tear, a region having a smaller thickness, etc.), that can facilitate easier removal of the magnet from the tether **112**. In this way, the magnet **114** can, after being removed from the patient while still coupled to the tether **112**, be subsequently removed from the tether **112** to ensure that the magnet **114** has been removed from the patient. In addition, the magnet **114** having been removed, can be cleaned (e.g., autoclaved), recycled, etc., which can be advantageous when the magnet **114** is a rare-earth magnet (e.g., a magnet that is costly).

[0047] In some embodiments, the tether **112** (e.g., in an uncoiled state) can be longer than the channel **120**. In some configurations, the tether **112** can transition from a compacted state (e.g., a coiled state) to an unfurled state. In this way, as the magnet **114** is moved, the tether **112** transitions from the compacted state to the unfurled state, without moving the slide **116** substantially (or at all). In some cases, although not shown in FIG. **1**, the slide **116** can be in contact with the stop **118** before moving the magnet **114** and unraveling of the tether **112**. In this way, as the magnet **114** is moved (e.g., pulled), the magnet **114** unravels the tether **112** until the tether **112** becomes completely unraveled. At this point, further pulling of the magnet **114** can cause the tether **112** to be tensilely loaded because the slide **116** is blocked by the stop **118**. This loading of the tether **112** can be felt by the practitioner, which can provide a tactile response for the practitioner to stop pulling the magnet **114**.

[0048] In some embodiments, the slide **116** and the magnet **114** can be replaced. In this case, the magnet **114** can be positioned where the slide **116** is in FIG. **1**, and the slide **116** can be positioned where the magnet **114** is in FIG. **1**. Thus, the magnet **114** can be positioned closer to the proximal end **104** of the endotracheal tube **102** than the slide **116**. Correspondingly, the mechanical stop **118** can be positioned at the opposing end of the channel **120** (e.g., so that the mechanical stop **118** is closer to the proximal end **104** than the distal end **106** of the endotracheal tube **102**). In this scenario, including when the magnet **114** is pulled (e.g., by another magnet) in a first direction, the slide **116** is also pulled in the first direction, until the slide **116** reaches and contacts the stop **118**. At this point, further puling of the magnet **114** will be blocked due to the stop **118** contacting and thus blocking movement of the slide **116** in the direction of the pulling. For example, as the magnet **114** is pulled, the tether **112** passes through (or by) the stop **118** (e.g., due to the size of tether being smaller than the stop **118**), however, the slide **116** is larger than the stop **118**, and thus the stop **118** is unable to move past the stop **118** in the pulling direction.

[0049] FIG. **2** shows a probe **150** that is configured to retrieve the magnet **114** with the probe **150** being depicted in a straight configuration, while FIG. **3** shows the probe **150** in a curved configuration. The probe **150** can include a body **152**, and a magnet **154** coupled to the body **152**. As shown in FIG. **2**, the magnet **154** can be coupled to a distal end of the body **152**, integrated within the body **152** of the probe **150**, etc. The magnet **154** can be implemented in a similar manner as the magnet **114** of the endotracheal tube **100**. For example, the magnet **154** can be a permanent magnet (e.g., a rare-earth magnet), and can be a single magnet, or multiple individual magnets coupled together. For example, in some cases, the magnet **154** can have multiple pairs of opposite poles (e.g., a north pole and a south pole), with each pair of poles being oriented in the same direction. In some cases, the magnet **154** can define an axis that extends through both poles of the magnet **154**, and the axis can be substantially (i.e., deviating by less than 10 percent from) parallel, or perpendicular to a length of the tube **102**. In some cases, the axis being substantially parallel to a longitudinal axis **156** of the body **152** can be advantageous in that the pole of the magnet **154** can be more in alignment with movement of the probe **150**, which can facilitate better coupling between the magnet **154** and the magnet **114**.

[0050] FIG. **3** shows the distal end of the body **152** including the magnet **154** being oriented away from the proximal end of the body **152** (e.g., the distal end and the proximal end of the body **152** are out of alignment with each other). The probe **150** is configured to curve to be oriented in different ways, so as to traverse around different structures within the nasal cavity, the pharynx, and the trachea. For example, the body **152** can be formed out of a flexible material (e.g., an elastic material) so that the body can curve to accommodate the different curved paths.

[0051] FIG. **4** shows a schematic illustration of an endotracheal tube **200**, which can be similar to the endotracheal tube **100**. Thus, the previous description of the endotracheal tube **100** also pertains to the endotracheal tube **200**. Similarly to the endotracheal tube **100**, the endotracheal tube **200** can include a tube **202** having a proximal end **204** and a distal end **206** opposite the proximal end **204**, an inflatable cuff **208**, a pilot balloon **210** in fluid communication with the inflatable cuff **208**, a tether **212**, and a magnet **214**. In addition, the endotracheal tube **200** can include a pouch **216** that can contain the tether **212** and the magnet **214**. As shown in FIG. **4**, the tether **212** can be coupled to the tube **202**, and the tether **212** can be coupled to the magnet **214**. For example, one end of the tether **212** can be coupled to the tube **202** (e.g., using a fastener, such as an adhesive), and the other end of the tether **212** can be coupled to the magnet **214**. Regardless of the configuration, the tether **212** can be coupled to the tube **202** and the magnet **214** can be coupled to the tether **212** so that the magnet **214** is separated from the tube **202**.

[0052] In some cases, the pouch **216** can be coupled to the tube **202** or the pouch **216** can be integrally formed with the tube **202** (e.g., the pouch **216** and the tube **202** being formed of the same material and being a single monolithic component). Regardless, the tether **212** with the magnet **214** coupled thereto can be positioned within the internal volume of the pouch **216** so that the pouch partially (or entirely) encloses the magnet **214**. In some cases, the pouch **216** can include a region of a material weakness **218**, which can extend partially (or entirely) around the pouch **216**. In some cases, the region of material weakness **218** can be a perforation, a tear, etc., and can generally facilitate removal of the magnet **214** from the pouch **216** and unraveling of the tether **212**. For example, the tether **212** can be in a compact state (e.g., a coiled state), when, for example, the tether **212** and the magnet **214** are positioned within the pouch **216**. Once the magnet **154** of the probe reaches the magnet **214** and the magnets **154**, **214** are magnetically coupled together, the probe **150** can be retracted, which forces the magnet **214** against the wall of the pouch **216** thereby tearing the wall of the pouch **216** at the region of material weakness **218**. At this point, the magnet **214** can be pulled away, via the magnet **154** of the probe **150**, to unravel the tether **212** from the compact state (e.g., the coiled state) and move the magnet **214** (and a portion of the tether **212**) out of the patient. In some embodiments, while the pouch **216** has been described as enclosing the magnet **214** and tether **212**, in other configurations, the pouch **216** can have a hole. In this case, the magnet **214** can be advanced through the hole, which can prevent needing to tear the pouch **216**.
[0053] In some embodiments, the endotracheal tube **200** can include a bar (not shown) that is coupled to the magnet **214** and the tube **202** so that the bar is positioned between the magnet **214** and the tube **202**. The bar can include a region of material weakness (e.g., a portion of the bar being thinner than another region of the bar), which can facilitate breakage of the bar at the region of material weakness. For example, after the magnets **154**, **214** are magnetically coupled together, the magnet **214** can be retracted (e.g., pulled) until the bar breaks in two (at the region of material weakness) thereby releasing the magnet **214** from the tube **202**.
[0054] FIG. **5** shows a schematic illustration of the endotracheal tube **100** deployed within the patient, which can be applicable to the other endotracheal tubes described herein (e.g., the endotracheal tube **200**). As shown in FIG. **5**, the distal end **106** of the tube **102** and the inflatable cuff **108** are positioned within the trachea of the patient, while the proximal end **104** of the tube **102** is positioned outside of the patient. For example, the distal end **106** of the tube **102** can be inserted into the mouth of the patient, through the mouth of the patient, and into the trachea. Correspondingly, with the distal end **106** of the tube **102** positioned within the trachea, the proximal end **104** of the tube **102** can be positioned outside of the patient. As shown in FIG. **5**, the tether **112** extends through the pharynx and through the nasal cavity so that a portion of the tether **112** is positioned outside of the patient. For example, after the magnet **154** of the probe **150** magnetically couples with the magnet **114**, the probe **150** is retreated thereby advancing the magnet **114** through the pharynx, through the nasal cavity, and out the nasal cavity. In some cases, and as described above, the stop **118** blocks further advancement of the slide **116**, which provides an anchor point for the tether **112** on the tube **102**. In some embodiments, once the magnet **114** (and a portion of the tether **112**) are positioned outside of the patient, the magnet **114** can be removed, and the tether **112** (which can sometimes include the magnet **114**) can be coupled to an anchor **122** that is external to the patient. In some embodiments, the anchor **122** can include a clip that couples the tube **102** and the tether **112** together, that couples the tube **102**, the tether **112**, and a bridle tubing (not shown in FIG. **5**) together, etc. In some configurations, the anchor **122** can include a tube holder (not shown in FIG. **5**) that is secured around the head of the patient, and which couples to the tube **102**. In this case, the tether **112** can also be secured to the tube holder.
[0055] As shown in FIG. **5**, with a portion of the tether **112** positioned outside of the patient (and coupled to the anchor **122**), the soft palate of the patient is positioned between the tube **102** and the tether **112**. Accordingly, when the proximal end **104** of the tube **102** is advanced away from the patient, the tether **112** contacts the soft palate thereby blocking further advancement of the

proximal end of the tube away from the patient. In this way, the tether **112** can help to ensure that the distal end **106** of the tube **102** is not undesirably removed from the trachea of the patient, by physically blocking the tube **102**, via the tether **112**. In addition, contacting of the soft palate with the tether **112** can be painful for the patient, and thus the patient is less likely to continue pulling the tube **102**, which would cause additional pain for the patient. Thus, the tether **112** can act as a deterrent for pulling the tube **102** out from the patient.

[0056] In some embodiments the endotracheal tube may be configured so that an external portion separates from the internal portion when a force is applied to the external portion, thereby preventing the internal portion from being removed (e.g., by a patient) at an inappropriate time. Accordingly, FIG. **6** shows a schematic illustration of an endotracheal tube system **300** with adapters separated from each other, while FIG. **7** shows a schematic illustration of an endotracheal tube system **300** with the adapters coupled together. The endotracheal tube system **300** can include a fluid source **302** (e.g., a ventilator), an endotracheal tube **304** including an adapter **306** (e.g., coupled to the proximal end of the tube of the endotracheal tube **304**), and an adapter **308** that is in fluid communication with the fluid source **302**. The adapter **306** is configured to be removably coupled to the adapter **308** (and vice versa). For example, the adapters **306**, **308** can be magnetically coupled together, can be a snap-fit connection, can include a hook and loop fastener (e.g., one of the adapters including a hook fastener and the other adapter including a loop fastener), etc. In some cases, the adapters **306**, **308** being magnetically coupled together can be advantageous in that not only can the magnets of the adapters **306**, **308** can be configured to provide a relatively small coupling force (e.g., easily decoupled from each other the advantages of which are described below), but the magnets of the adapters **306**, **308** can also guide a user to easily reconnect the adapters **306**, **308** together (e.g., in the case bodily fluids including blood, vomit, etc., that obstruct the view of the adapters **306**, **308** in the mouth of the patient). For example, the adapters **306**, **308** can each include a respective magnet **310**, **312**, with the magnets **310**, **312** having opposing poles that face each other. In other words, a north pole of the magnet **310** can face the south pole of the magnet **312** (or vice versa).

[0057] In some configurations, the adapter **306** can include an extension **314** that extends away from the tube of the endotracheal tube **304** (e.g., a distal end of the endotracheal tube **304**), and the magnet **310**. In some cases, the extension **314** can be inserted into the adapter **308** (e.g., so that the magnet **312** surrounds the extension **314**), when the adapters **306**, **308** are coupled together. Which can not only guide coupling of the adapters **306**, **308** together, but can also help to better ensure a fluid tight seal between the adapters **306**, **308**. While the adapter **306** has been illustrated as including the extension **314**, in other cases, the adapter **308** can include the extension **314**, in which case the extension **314** is inserted into the adapter **306**.

[0058] As shown in FIG. **7**, the adapters **306**, **308** are coupled together with the magnets **310**, **312** being magnetically coupled together (e.g., and the magnets **310**, **312** in contact with each other). In some embodiments, and advantageously, the force needed to decouple the adapters **306**, **308** can be less than the force required to dislodge the inflatable cuff (or other balloon) that is positioned within the trachea of the patient. In this way, with the adapters **306**, **308** coupled together, pulling of the adapter **308** away from the adapter **306** decouples the adapters **306**, **308**, rather than pulling the endotracheal tube **304**, which would pull the distal end of the endotracheal tube **304** out of the trachea. Thus, the endotracheal tube **304** advantageously continues to be positioned within the trachea, even when the adapters **306**, **308** are decoupled. In addition, the adapter **306** can be positioned within the mouth of the patient when the distal end of the endotracheal tube **304** is within the trachea, so that only the adapter **308** is positioned outside of the patient. In this way, when the patient pulls the adapter **308** (e.g., which is more likely to happen because the adapter **308** is more accessible to the patient), the adapters **306**, **308** simply decouple and the distal end of the endotracheal tube **304** (e.g., including the inflatable cuff) does not move substantially away from the position of the distal end of the tube within the trachea prior to pulling the second adapter.

Thus, when the adapters **306**, **308** are decoupled, the distal end of the endotracheal tube **304** advantageously remains in the trachea of the patient.
[0059] In some embodiments, the endotracheal tube **304** can include a balloon **316** that can extend partially (or entirely) around the tube of the endotracheal tube **304**. The balloon **316** can be selectively inflated and deflated, via a fluid source. For example, when the endotracheal tube **304** is deployed within the patient (e.g., the endotracheal tube **304** is positioned within the trachea of the subject), the balloon **316** can be positioned above the vocal cords. In this way, the balloon **316** can be inflated to contact a wall of the trachea thereby preventing the distal end of the endotracheal tube **302** from falling down further into the trachea, if, for example, the adapters **306**, **308** decouple from each other (e.g., by the patient pulling on the adapter **308**, or a component coupled to the adapter **308**, such as other tubing).
[0060] FIG. **8** shows a schematic illustration of an endotracheal tube system **350**, which relates to the other endotracheal tube systems described herein. Thus, the endotracheal tube systems described herein pertain to the endotracheal tube system **350** (and vice versa). The endotracheal tube system **350** can include an endotracheal tube **352** including an adapter **354** (e.g., coupled to a proximal end of the endotracheal tube **352**), and an anchor **356**. The endotracheal tube **352** can be coupled to (e.g., removably coupled to) the anchor **356**, and in particular, the adapter **354** can be coupled to (e.g., removably coupled to) the anchor **356**. For example, including when the adapter **354** includes a magnet **358**, the anchor **356** can also include a magnet that magnetically couples to the magnet **358**.
[0061] The anchor **356** can be implemented in different ways. For example, the anchor **356** can be positioned outside of the patient and can be coupled to the outside portion of the patient. For example, the anchor can be fastened around an anatomical structure of the patient, including, for example, the head of the patient. In some cases, the anchor **356** can include a tube holder (e.g., a Hollister tube holder, an endotracheal tube fastener, etc.) that couples to the patient (e.g., by adhering to an external surface of the patient). In some embodiments, the anchor **356** can include a clip that couples together the endotracheal tube **352** (including the tube of the endotracheal tube **352**, a tether of the endotracheal tube **352**, etc.), a bridle tube, etc. In some cases, when the anchor **356** includes a clip and a tube holder, the clip can couple the tube holder, the tube of the endotracheal tube **352**, the tether of the endotracheal tube **352**, and a bridle tube together. In some cases, the clip can be a frictional clip (e.g., a snap-fit clip), a magnetic clip, etc. In some configurations, when the clip is a frictional clip, the clip can better secure the components together.
[0062] FIG. **9** shows a schematic illustration of an endotracheal tube system **400**, which relates to the other endotracheal tube systems described herein. Thus, the endotracheal tube systems described herein pertain to the endotracheal tube system **400** (and vice versa). The endotracheal tube system **400** can include an endotracheal tube **402**, a tube holder **404**, and a ventilator tube **406**. The endotracheal tube **402** can include a tube **408**, and an adapter **410** coupled to the tube **408**. The tube holder **404** can be positioned outside of the patient and can be coupled to an exterior surface of the patient. For example, the tube holder **404** can include one or more straps **412** that can extend around the head of a patient to secure the tube holder **404** to the head of the patient. In some cases, the tube holder **404** can include a bracket **414** that can be coupled to the one or more straps **412** and can extend around a portion of the face of the patient. In some cases, the bracket **414** can be positioned below a nose of the patient, and above the mouth of the patient (e.g., when the tube holder **404** is secured to the patient). As shown in FIG. **9**, the tube holder **404** can be coupled to the endotracheal tube **402**, which can include the tube holder **404** being removably coupled to the endotracheal tube **402** (e.g., being snap fitted, being magnetically coupled, etc.). For example, the tube holder **404** can include an arm **416** that can be coupled to the bracket **414** at one end of the arm **416**, and can be coupled to the adapter **410** of the endotracheal tube **402** at the opposing end of the arm **416**. In some cases, the arm **416** can be removably coupled to the adapter **410** (e.g., the arm **416** being magnetically coupled to the arm **416**). In this way, after the endotracheal tube **402** has

been deployed, the tube holder **404** can be secured around the patient, and the arm **416** of the tube holder **404** can be coupled to the adapter **410** of the endotracheal tube **402** to secure the endotracheal tube **402** (e.g., to prevent inadvertent pulling out of the endotracheal tube **402**).

[0063] As shown in FIG. **9**, the endotracheal tube **402** can include a tether **418**, and a magnet **420** that is coupled to the tether **418**. In some cases, after the endotracheal tube **402** has been deployed and the magnet **420** has been advanced through and out the nasal passage of the patient, the magnet **420** can be coupled to the adapter **410** (e.g., with the magnet **420** being removably coupled to the adapter **410**). In this way, the tether **418** can be prevented from being retreated back through the nasal passage of the patient. In other configurations, the tether **418** can be coupled to the endotracheal tube **402** (e.g., can be tied around the adapter **410**). In some embodiments, the ventilator tube **406** can include an adapter **422** that can be removably coupled to the adapter **410** of the endotracheal tube **402**. For example, after the endotracheal tube **402** has been deployed, the magnet **420** retrieved, and the tube holder **404** secured to the patient, the adapter **422** of the ventilator tube **406** can be coupled to the adapter **410** of the endotracheal tube **402**. In this way, with the adapters **410**, **422** coupled together, the ventilator tube **406** can provide gas (e.g., air) to and through the endotracheal tube **402**, and the if the patient inadvertently pulls on the ventilator tube **406**, the ventilator tube **406** decouples from the endotracheal tube **402** at the adapters **410**, **422**, without the endotracheal tube **402** being pulled with the ventilator tube **406**. This can be advantageous in that the risk for inadvertent pulling out of the endotracheal tube **402** can be significantly mitigated.

[0064] The present disclosure has described one or more preferred embodiments, and it should be appreciated that many equivalents, alternatives, variations, and modifications, aside from those expressly stated, are possible and within the scope of the invention.

[0065] It is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

[0066] As used herein, unless otherwise limited or defined, discussion of particular directions is provided by example only, with regard to particular embodiments or relevant illustrations. For example, discussion of “top,” “front,” or “back” features is generally intended as a description only of the orientation of such features relative to a reference frame of a particular example or illustration. Correspondingly, for example, a “top” feature may sometimes be disposed below a “bottom” feature (and so on), in some arrangements or embodiments. Further, references to particular rotational or other movements (e.g., counterclockwise rotation) is generally intended as a description only of movement relative a reference frame of a particular example of illustration.

[0067] Certain operations of methods according to the disclosure, or of systems executing those methods, may be represented schematically in the figures or otherwise discussed herein. Unless otherwise specified or limited, representation in the figures of particular operations in particular spatial order may not necessarily require those operations to be executed in a particular sequence corresponding to the particular spatial order. Correspondingly, certain operations represented in the figures, or otherwise disclosed herein, can be executed in different orders than are expressly illustrated or described, as appropriate for particular embodiments of the disclosure. Further, in some embodiments, certain operations can be executed in parallel, including by dedicated parallel processing devices, or separate computing devices configured to interoperate as part of a large system.

[0068] As used herein in the context of computer implementation, unless otherwise specified or limited, the terms “component,” “system,” “module,” and the like are intended to encompass part or all of computer-related systems that include hardware, software, a combination of hardware and software, or software in execution. For example, a component may be, but is not limited to being, a processor device, a process being executed (or executable) by a processor device, an object, an executable, a thread of execution, a computer program, or a computer. By way of illustration, both an application running on a computer and the computer can be a component. One or more components (or system, module, and so on) may reside within a process or thread of execution, may be localized on one computer, may be distributed between two or more computers or other processor devices, or may be included within another component (or system, module, and so on).

[0069] In some implementations, devices or systems disclosed herein can be utilized or installed using methods embodying aspects of the disclosure. Correspondingly, description herein of particular features, capabilities, or intended purposes of a device or system is generally intended to inherently include disclosure of a method of using such features for the intended purposes, a method of implementing such capabilities, and a method of installing disclosed (or otherwise known) components to support these purposes or capabilities. Similarly, unless otherwise indicated or limited, discussion herein of any method of manufacturing or using a particular device or system, including installing the device or system, is intended to inherently include disclosure, as embodiments of the disclosure, of the utilized features and implemented capabilities of such device or system.

[0070] As used herein, unless otherwise defined or limited, ordinal numbers are used herein for convenience of reference based generally on the order in which particular components are presented for the relevant part of the disclosure. In this regard, for example, designations such as “first,” “second,” etc., generally indicate only the order in which the relevant component is introduced for discussion and generally do not indicate or require a particular spatial arrangement, functional or structural primacy or order.

[0071] As used herein, unless otherwise defined or limited, directional terms are used for convenience of reference for discussion of particular figures or examples. For example, references to downward (or other) directions or top (or other) positions may be used to discuss aspects of a particular example or figure, but do not necessarily require similar orientation or geometry in all installations or configurations.

[0072] This discussion is presented to enable a person skilled in the art to make and use embodiments of the disclosure. Various modifications to the illustrated examples will be readily apparent to those skilled in the art, and the generic principles herein can be applied to other examples and applications without departing from the principles disclosed herein. Thus, embodiments of the disclosure are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein and the claims below. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected examples and are not intended to limit the scope of the disclosure. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of the disclosure.

[0073] Various features and advantages of the disclosure are set forth in the following claims.

Claims

1. An endotracheal tube comprising: a tube having a distal end that is configured to be inserted into the trachea of a patient and a proximal end that is configured to be positioned within the mouth of the patient or outside of the patient; a tether having a first end coupled to the tube; and a magnet coupled to the tether, the magnet being configured to be advanced through the pharynx, through and out the nasal cavity thereby positioning the soft palate of the patient between the tube and the tether.

2. The endotracheal tube of claim 1, wherein when the proximal end of the tube is advanced away from the patient, the tether contacts the soft palate to block further advancement of the proximal end of the tube away from the patient.

3. The endotracheal tube of claim 1, further comprising: a channel that extends along the tube; a slide coupled to a second end of the tether opposite the first end; and a stop coupled to the tube, wherein the magnet is configured to be moved along the channel in a first direction towards the proximal end of the tube until the slide contacts the stop thereby blocking further advancement of the magnet in the first direction.

4. The endotracheal tube of claim 1, wherein a second end of the tether opposite the first end is configured to be coupled to an anchor that is external to the patient thereby fixing the position of the endotracheal tube within the patient.

5. The endotracheal tube of claim 4, wherein the anchor wraps around the head of the patient, and wherein a portion of the tube is coupled to the anchor.

6. The endotracheal tube of claim 1, wherein the magnet is removably coupled to the tether.

7. The endotracheal tube of claim 1, further comprising a first adapter coupled to the proximal end of the tube, wherein a second adapter of a conduit in fluid communication with a ventilator is configured to be removably coupled to the first adapter, and wherein the first adapter is configured to be positioned within the mouth of the patient when the distal end of the tube is placed in the trachea of the patient.

8. The endotracheal tube of claim 7, wherein when the second adapter is pulled away from the first adapter of the tube, the second adapter decouples from the first adapter.

9. The endotracheal tube of claim 7, wherein the magnet is a first magnet, wherein the first adapter comprises a second magnet, wherein the second adapter comprises a third magnet, and wherein the first adapter magnetically couples to the second adapter, via the interaction between the second magnet and the third magnet.

10. The endotracheal tube of claim 7, further comprising an inflatable cuff coupled to the tube and configured to be positioned within the trachea of the patient, and wherein when the first adapter is coupled to second adapter, and the second adapter is pulled away from the first adapter, the second adapter decouples from the first adapter with the distal end of the tube remaining in the trachea.

11. The endotracheal tube of claim 1, wherein the magnet is a first magnet, and wherein a probe with a second magnet coupled thereto is configured to be inserted into the nasal cavity, through the nasal cavity, into the pharynx to be magnetically coupled to the first magnet, and wherein with the second magnet magnetically coupled to the first magnet, the probe is configured to be retreated back through and out the nasal cavity.

12. A method for deploying an endotracheal tube comprising: inserting a distal end of a tube into the trachea of a patient; positioning a proximal end of the tube within the mouth of the patient or outside of the patient, a first end of a tether being coupled to the tube, and a magnet being coupled to the tether; and advancing the magnet through the pharynx of the patient, through and out the nasal cavity such that the soft palate of the patient is positioned between the tube and the tether.

13. The method of claim 12, wherein, when the proximal end of the tube is advanced away from the patient, the method further comprises: contacting the soft palate with the tether to block further advancement of the proximal end of the tube away from the patient.

14. The method of claim 12, further comprising: providing a channel that extends along the tube, wherein a slide is coupled to a second end of the tether opposite the first end and a stop is coupled to the tube, and moving the magnet along the channel in a first direction towards the proximal end of the tube until the slide contacts the stop to block further advancement of the magnet in the first direction.

15. The method of claim 12, further comprising: coupling a second end of the tether opposite the first end to an anchor that is external to the patient to fix the position of the endotracheal tube within the patient.

16. The method of claim 15, further comprising: wrapping the anchor around the head of the patient, and coupling a portion of the tube to the anchor.

17. The method of claim 12, wherein the magnet is removably coupled to the tether.

18. The method of claim 12, further comprising providing a first adapter coupled to the proximal end of the tube, wherein a second adapter of a conduit in fluid communication with a ventilator is configured to be removably coupled to the first adapter, and positioning the first adapter within the mouth of the patient when the distal end of the tube is placed in the trachea of the patient.

19. The method of claim 18, wherein, when the second adapter is pulled away from the first adapter of the tube, the method further comprises: decoupling the second adapter from the first adapter.

20. The method of claim 18, wherein the magnet is a first magnet, wherein the first adapter comprises a second magnet, wherein the second adapter comprises a third magnet, and wherein the method further comprises: magnetically coupling the first adapter to the second adapter based on the interaction between the second magnet and the third magnet.

21. The method of claim 18, further comprising: providing an inflatable cuff coupled to the tube, positioning the inflatable cuff within the trachea of the patient, coupling the first adapter to the second adapter, and decoupling, when the second adapter is pulled away from the first adapter, the second adapter from the first adapter, wherein the distal end of the tube remains in the trachea.

22. The method of claim 12, wherein the magnet is a first magnet, and wherein the method further comprises: inserting a probe with a second magnet coupled thereto into the nasal cavity, through the nasal cavity, and into the pharynx, magnetically coupling the second magnet to the first magnet, and retreating the probe with the second magnet magnetically coupled to the first magnet back through and out the nasal cavity.